

ZoneWatch

User Guide

Turn any camera into a smart occupancy sensor.

Version 2.3 — covers Windows, Linux, macOS, Raspberry Pi and Docker.

1. What is ZoneWatch?

ZoneWatch watches your cameras with built-in artificial intelligence and tells you whether the areas you care about — a parking spot, a driveway, a desk, a machine — are **FREE** or **OCCUPIED**, in real time, with history and notifications. Everything runs on your own computer: no cloud account, no subscription, and your video never leaves your network.

How it works

1. You connect a camera (IP camera, ESP32-CAM, USB webcam or a video file).
2. You draw one or more **zones** (polygons) directly on the camera image.
3. Several times per second, the AI looks at the full image and finds objects (cars, people, dogs...).
4. If a matching object stands inside a zone, the zone becomes OCCUPIED; when it leaves, the zone becomes FREE. Every change is recorded with a snapshot and can trigger notifications.

Detection is real object recognition — not motion or pixel-change detection. A car parked in a zone is detected even if it never moves, and shadows or trees moving in the wind do not cause false alarms.

2. Installation

2.1 Windows

1. Unzip the ZoneWatch folder anywhere convenient — Documents, Desktop or C:\ZoneWatch. **Avoid C:\Program Files** (Windows restricts writing there and the app stores its data next to itself).
2. Open the folder and double-click **zonewatch.exe**.
3. The first time, Windows SmartScreen may show “Windows protected your PC”. Click **More info -> Run anyway** — this is normal for independent software.
4. A console window opens (that is the server — keep it open). Open your browser at **http://localhost:8484**.

2.2 Linux and Raspberry Pi

Unzip, open a terminal in the folder, then:

```
chmod +x zonewatch # first time only
./zonewatch
```

Open <http://localhost:8484> in a browser. To access from another device on your network, use <http://<machine-ip>:8484>.

2.3 macOS

Unzip and double-click **zonewatch**. macOS will warn about an unidentified developer the first time: right-click the file -> **Open** -> Open. Then browse to <http://localhost:8484>.

2.4 First start

On first start ZoneWatch downloads its AI model (~19 MB) — the only time it needs internet. Then create your administrator account (username + password of at least 8 characters). These credentials protect the app on your network; there is no way to recover a forgotten password other than resetting the database (see Troubleshooting), so store it safely.

Where your data lives

Everything is stored inside the ZoneWatch folder: **data/** (database, AI models, event snapshots) and **config/** (configuration files). Back up these two folders and you can restore or move your whole installation to another machine.

3. Five-minute quick start

Step	What to do
1	Sign in, go to Cameras -> + Add camera . Pick the type, paste the address, click Test connection , then Add .
2	Go to Zones . Click on the camera image to place corner points around the area you care about; double-click (or “Finish shape”) to close it.
3	Name the zone, choose what to watch for (e.g. the Vehicles preset), click Create zone .
4	Open the Dashboard — your zone shows FREE or OCCUPIED live on the video.
5	Check Events after a few minutes to see history, snapshots and statistics.

Tip: No camera at hand? Settings -> Detection -> enable **Demo mode**, then add a camera of type **Video file** pointing at any video on your disk. Fake detections let you explore everything.

4. Cameras

4.1 Supported sources

Type	Use for	Address example
RTSP	Almost every IP camera (Hikvision, Dahua, Reolink, TP-Link...)	rtsp://user:pass@192.168.1.10:554/stream1
HTTP / MJPEG	ESP32-CAM, some older cameras, phone camera apps	http://192.168.1.20:81/stream
USB webcam	Webcams plugged into this computer	0 (first webcam), 1 (second)...
Video file	Testing, demos, analysing recordings	C:\videos\parking.mp4 (loops forever)

Find your camera's RTSP address in its app/manual or search “<brand> RTSP url”. Phones can act as IP cameras with free apps (e.g. “IP Webcam” on Android -> http://<phone-ip>:8080/video).

4.2 Camera settings explained

Field	Meaning	Recommended
Name	Label shown everywhere	“Front parking”, “Workshop”...

Field	Meaning	Recommended
Type / Source	See table above	Use Test connection before saving
Detect FPS	How many times per second the AI analyses this camera. Higher = faster reactions, more CPU.	Leave empty (uses the global setting). Parking: 1–2 is plenty.
Downscale width for AI	Shrinks the image before analysis to save CPU. 0 = off.	960 for 4K cameras, 0 otherwise
Enabled	Untick to pause a camera without deleting it	—

4.3 Managing cameras

Edit and **Delete** buttons are on each row of the Cameras page. Deleting a camera also deletes its zones and their event history (a confirmation is asked). The status column shows online / connecting / error with the measured stream frame rate.

5. Zones

5.1 Drawing

Zones page -> choose the camera (top right) -> click on the image to drop corner points -> **double-click** or press “Finish shape” to close the polygon (minimum 3 points). “Clear drawing” discards an unfinished shape. “Refresh image” fetches a new snapshot.

Important: Draw zones on the **ground** area where objects stand — not around roofs or upper bodies. ZoneWatch anchors each object near the bottom-centre of its box (where wheels/feet touch the ground), which makes detection reliable with angled cameras.

5.2 Zone settings explained

Field	Meaning	Recommended
Name	Shown on video, dashboard, events, Home Assistant	“Spot 1”, “Driveway”...
Watch for	Which object types trigger this zone. Presets: Vehicles, People, People & Vehicles, Animals, Anything. Nothing ticked = any object.	Parking: Vehicles. Security: People.
Min confidence	How sure the AI must be (0–1). Lower = catches more, more false alarms. Empty = global setting.	Far/small objects: 0.30–0.40
Debounce frames	How many consecutive analyses must agree before the state flips. Filters out cars driving past.	3 (≈1.5 s at 2 det/s). Busy street: 5.
Color	Zone colour in the editor and live view	—
Enabled	Pause the zone without deleting	—
Send notifications	Whether this zone fires webhooks	—

5.3 Managing zones

Click a zone in the list to highlight it; **Edit** opens its settings (including Delete). Multiple zones per camera are fine — e.g. one small zone per parking spot plus one large counting zone across the whole row.

One zone per spot, or one big zone?

A **small zone per parking spot** answers “is spot 3 free?” — each spot gets its own state, history and sensor. A **large zone** answers “how many cars are in this area?” (the count shows as ×N). For parking, per-spot zones are what you want; add a big zone only for area statistics.

6. Dashboard

The dashboard shows live annotated video for every camera: blue boxes are detected objects with confidence %, coloured polygons are your zones (green = free, red = occupied), and each camera card lists its zones with their current state and how long since the last change. The camera selector (top right) focuses the page and the summary tiles on one camera. Everything refreshes automatically.

7. Events & statistics

Every state change is recorded: time, zone, camera, occupied/vacated, the object type, how long the zone had been occupied, and a snapshot image (click to enlarge). Filter by camera, zone, event type and time window; pages of 50 events.

The cards at the top show per zone: **occupancy %** over the selected window, number of **arrivals**, current state, and an **arrivals-by-hour-of-day** chart — perfect for spotting busy hours of a shop, parking or meeting room.

Retention: events are kept 90 days and snapshots 30 days by default (changeable in Settings).

8. Settings reference

8.1 Detection

Setting	Meaning
Model	yolox_nano : fastest, lowest accuracy — very old PCs / Raspberry Pi. yolox_tiny : recommended balance. yolox_s : most accurate, best for small/far objects (uses more CPU). Models download automatically on first use.
Confidence	Global minimum certainty (0–1) for a detection to count. Default 0.45. Zones can override it.
Detections/sec per camera	Global analysis rate. 1–2 is enough for parking; 3–5 for fast arrival alerts. Biggest lever on CPU usage.
Demo mode	Generates fake detections so you can explore without a camera or model. Turn OFF for real use.
GPU acceleration	Toggle appears active when a GPU runtime is detected; the hint line shows the processor actually in use. See 8.5.

8.2 Home Assistant / MQTT

Setting	Meaning
Enable MQTT	Master switch for the integration
Broker host / Port	Address of your MQTT broker — typically your Home Assistant IP with the Mosquitto add-on, port 1883
Username / Password	An MQTT user (create one in Home Assistant -> People or in the Mosquitto add-on)
Home Assistant auto-discovery	When on, every zone automatically appears in HA as a device with an occupancy binary_sensor and a count sensor. No YAML needed.

Example HA automation: trigger on *binary_sensor.spot_1_occupancy* turning **on** -> send a notification “Someone parked in your spot”.

8.3 Webhooks

Webhooks send an HTTP POST with JSON to any URL on every zone event — usable with Discord, Slack, ntfy.sh, Node-RED, IFTTT-style services or your own scripts. One URL per line; add a word after the URL to use it as a signing secret (an *X-Signature* HMAC-SHA256 header is then added). **Cooldown** limits how often a single zone may fire. Payload:

```
{"event": "occupied", "zone_id": 3, "zone": "Spot 1", "camera": "Front parking",  
"count": 1, "label": "car", "duration_seconds": null, "timestamp": 1760000000.0}
```

8.4 License

Version	Limits
Free (no key)	1 camera, 3 zones — all features otherwise
Personal	Unlimited cameras and zones
Business	Unlimited + commercial usage terms and priority support

Paste your key (starts with **ZWL-**) in Settings -> License -> Activate. Keys work offline — no internet needed. If a key has an expiry date and passes it, the app simply returns to the free limits; your cameras, zones and history are never deleted.

8.5 GPU acceleration

Platform	GPU support
Windows (NVIDIA / AMD / Intel)	Automatic — DirectML is bundled. “Dml” in the hint = GPU active.
macOS	Automatic — Apple GPU/Neural Engine via CoreML.
Linux with NVIDIA	Automatic when the CUDA runtime is present (the source install detects it).

Platform	GPU support
Raspberry Pi	CPU only — the Pi's GPU is not supported by any ML runtime (platform limitation). Use yolox_nano at 1–2 det/s.

If GPU initialisation ever fails, ZoneWatch falls back to CPU automatically — it never refuses to start.

8.6 API keys

Create keys for integrations (read-only or admin). Send the key in the *X-API-Key* header:

```
curl -H "X-API-Key: zw_..." http://localhost:8484/api/state
curl -H "X-API-Key: zw_..." "http://localhost:8484/api/events?limit=10"
curl -H "X-API-Key: zw_..." "http://localhost:8484/api/stats?hours=24"
```

The key is shown once at creation — store it. Revoke keys anytime from the same card.

8.7 Account & data

Change password (signs out all other sessions). **Export backup** downloads a JSON with cameras, zones and settings; **Import backup** restores it — handy for moving to a new machine (snapshots/history are not included; copy the `data/` folder for a full clone).

9. What can ZoneWatch detect?

80 object types (COCO classes). The most reliable for zone monitoring are bolded:

Group	Objects
Vehicles	car, truck, bus, motorcycle, bicycle , boat, train, airplane
People	person
Animals	cat, dog , bird, horse, sheep, cow, elephant, bear, zebra, giraffe
Bags & items	backpack, handbag, suitcase, umbrella, tie, frisbee, sports ball, kite, skis, snowboard, skateboard, surfboard, tennis racket, baseball bat/glove
Food & kitchen	bottle, wine glass, cup, fork, knife, spoon, bowl, banana, apple, sandwich, orange, broccoli, carrot, hot dog, pizza, donut, cake
Furniture & home	chair, couch, potted plant, bed, dining table, toilet, tv, laptop, mouse, remote, keyboard, cell phone, microwave, oven, toaster, sink, refrigerator, book, clock, vase, scissors, teddy bear, hair drier, toothbrush
Street objects	traffic light, fire hydrant, stop sign, parking meter, bench

Accuracy depends on size in the image: an object should be at least roughly 1/20th of the image width to be detected reliably. For far scenes choose the yolox_s model and/or lower the zone confidence to ~0.35.

10. Recipes

Parking spot alert on your phone

Zone on the spot (Vehicles, debounce 3) -> Settings -> Webhooks -> add an ntfy.sh URL (e.g. <https://ntfy.sh/your-secret-topic>) -> install the free ntfy app and subscribe to the topic. Occupied/vacated alerts arrive as push notifications. With Home Assistant, use the auto-created `binary_sensor` instead.

Driveway arrival light (Home Assistant)

Zone across the driveway (Vehicles + People) -> HA automation: occupancy turns on AND sun below horizon -> turn on outdoor lights.

Meeting room / desk usage statistics

One zone per desk or table (People, debounce 5) -> Events page shows occupancy % and busiest hours per zone over any window.

Safety zone in a workshop

Zone around the dangerous machine area (People, confidence 0.5, debounce 1 for fast reaction, detections/sec 5) -> webhook to a local alarm or HA siren.

Pet on the counter

Zone on the kitchen counter (preset Animals, or just cat) -> webhook to ntfy -> photo evidence is in Events -> snapshots.

11. Performance & troubleshooting

Performance tuning

- Lower **Detections/sec** first — 1/s halves CPU vs 2/s with little practical loss.
- Use **yolox_nano** on weak machines, **yolox_s** only when accuracy demands it.
- Set camera **Downscale width** to 960 for 4K streams.
- Typical usage: a modern PC handles 4–8 cameras at 2 det/s on CPU alone.

Common problems

Problem	Solution
Camera shows “error” / no image	Click Edit -> Test connection. Check the address, username/password inside the URL, and that the camera is reachable from this machine (same network/VLAN). For RTSP try stream2 / substream paths.
Zone says FREE but a car is there	Demo mode still on? (Settings -> Detection). Object too small -> use yolox_s, lower zone confidence to 0.35, or move the camera closer. Zone drawn on the car's roof -> redraw on the ground.
False occupied flickers	Raise the zone's debounce frames (5–8) and/or confidence.
High CPU	Lower detections/sec; smaller model; downscale 4K cameras.
“No GPU detected”	GPU runtime missing — Windows: run install-gpu.bat (source install) or use a release build; the app still works fine on CPU.

Problem	Solution
Forgot admin password	Stop the app and delete the file data/zonewatch.db — this resets users AND cameras/zones (export a backup first if you can still sign in from another session).
Webhooks not arriving	Test the URL with curl first; check the zone has “Send notifications” ticked; check cooldown isn't suppressing them.
Model download fails	First start needs internet access to github.com; check firewall/proxy, then restart the app.

Accessing from other devices

The web interface is available at `http://<machine-ip>:8484` from any device on your network. For access over the internet, do NOT expose the port directly — use a VPN (WireGuard/Tailscale) or a reverse proxy with HTTPS.

12. Getting help

Questions, bug reports, camera compatibility reports and feature requests are handled on GitHub: open an issue at github.com/Vithusan21/zonewatch-releases. Please include your platform (Windows/Linux/macOS/Raspberry Pi), your camera model, and the last lines of the console window if something failed. Checking existing issues first often gives an instant answer.